

Inflation Inequality Across the Euro Area

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Household Inflation Exposure in the Euro Area

- How do relative-price shocks affect households with different consumption baskets?
- Monthly inflation indices by income, age and residence for 20 euro-area countries, excluding rents (041, 042), calibrated to HICP weights on matched products.
- Counterfactual price paths quantify how fiscal measures changed average inflation and income-group gaps.
- Household microdata reveal the dispersion concealed by group averages.

What Our Inflation Indices Capture

Channel	Mechanism	Covered here?
Consumption baskets	Different expenditure shares imply different inflation exposure	Yes
Nominal income	Wages, pensions and transfers adjust to inflation	No
Balance sheets	Unexpected inflation revalues nominal assets and debt; asset prices may also adjust	No

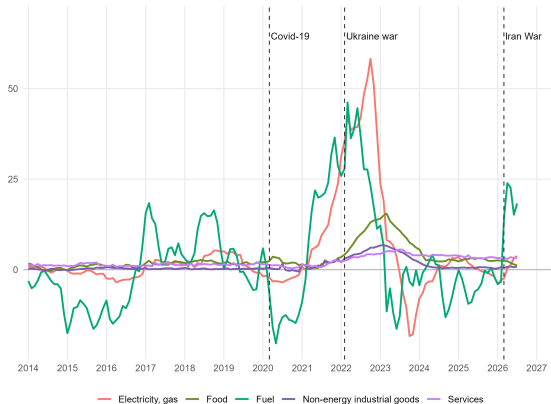
Different inflation rates do not by themselves rank households' welfare losses.

Main Findings, 2021–2023

- The annual Q1–Q5 inflation gap reached **1.2 percentage points in 2022**, with large differences across countries.
- Price policies lowered inflation by **1.4 points in 2022** and reduced the income gap by **0.5 point**.
- Household characteristics explain only a small part of inflation dispersion within countries.
- In 2022, scaled inflation exposure was **12.3% of income for Q1**, against **6.1% for Q5**, using 2020 expenditure-to-income ratios.

Energy and Food Drove Large Relative-Price Changes

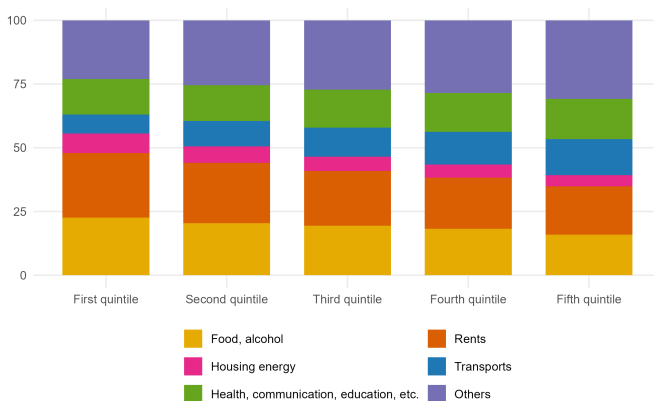
Euro area, year-on-year inflation (%).



Source: Eurostat HICP and HBS; authors' calculations.

Consumption Baskets Differ Across Income Quintiles

Shares of total expenditure including rents, 2020 (%).



Source: Eurostat HICP and HBS; authors' calculations.

An introductory example

Effect of a 40% energy price shock

Expenditure share	First quintile of income	Fifth quintile of income	Inflation gap (Q1–Q5)
Housing Energy	10%	6%	–
Other consumption	90%	94%	–
Inflation if housing energy prices rise by 40%	4.0%	2.4%	1.6 pp

Illustrative example; other prices unchanged, fixed expenditure weights.

Linking HICP Prices to Household Budget Surveys

- **Prices:** monthly HICP component indices and annual official product weights.
- **Baskets:** HBS expenditure profiles by income, age and residence, usually observed every five years.
- **Matching:** harmonized three-digit COICOP categories across 20 countries; national HBS sources supplement Italy.
- **Coverage:** exclude actual and imputed rents (041, 042), then renormalize. Within-product prices and substitution are unobserved.

Monthly and Annual Inflation Use Chained Indices

Within each year, prices are compared with the preceding December:

$$P_{g,y,m} = \sum_j \omega_{j,g,y} \frac{p_{j,y,m}}{p_{j,y-1,12}}, \quad I_{g,y,m} = I_{g,y-1,12} P_{g,y,m}.$$

- Monthly figures: year-on-year changes in $I_{g,y,m}$.
- Annual tables: percentage change in the annual-average index.
- Euro-area indices: aggregation with annual HICP country weights.

The annual Q1–Q5 gap is the difference between the two annual inflation rates, in percentage points.

Group Baskets Aggregate to Official HICP Weights

Let $s_{g,y}$ be group g 's share of total consumption and $\bar{\omega}_{j,y}^{\text{HICP}}$ the official product share on matched non-rent products.

$$\sum_j M_{j,g,y} = s_{g,y}, \quad \sum_g M_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}.$$

RAS adjusts the product-by-group expenditure matrix M to match both margins.

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}}, \quad \sum_g s_{g,y} \omega_{j,g,y} = \bar{\omega}_{j,y}^{\text{HICP}}.$$

The calibration uses expenditure shares, not population shares.

The Income Inflation Gap Peaked During the Energy Shock

ECB analysis reports similar findings using a slightly different methodology (Bobasu et al., 2023).



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Calculate and visualize inflation inequality with our R package:

github.com/JeremiMontornes/inflationinequality

National Income Gaps Differed in Magnitude and Sign

Q1–Q5 gap (pp)	2021	2022	2023
Latvia	-0.2	4.8	2.5
Estonia	0.5	4.4	1.3
Ireland	0.5	3.0	1.1
Italy	0.1	1.6	0.5
Germany	-0.4	1.4	0.6
France	0.3	0.2	0.2
Finland	-0.2	-0.9	0.4
Euro area	0.1	1.2	0.1

Household Energy and Food Widened the Income Gap

The distributional impact of inflation depends on which prices rise, as well as on the size of the aggregate shock.

Product group	Q1–Q5 inequality contribution		
	2021	2022	2023
Food and non-alcoholic beverages	0.1	0.7	0.7
Alcohol, tobacco and narcotics	0.1	0.1	0.2
Clothing and footwear	0.0	0.0	-0.1
Housing maintenance and furnishings	-0.1	-0.3	-0.2
Housing energy	0.4	1.8	0.1
Operation of personal transport equipment	-0.3	-0.4	0.0
Transport (excluding vehicle operation)	-0.1	-0.3	-0.2
Other	-0.1	-0.5	-0.5
Total	0.1	1.2	0.1

Source: Eurostat HICP and HBS; authors' calculations. Product rows are contributions to the Q1–Q5 annual inflation-rate gap, in percentage points.

Older Households Faced Higher Inflation



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

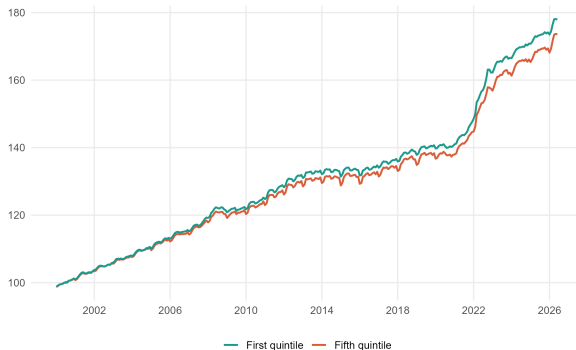
Rural Households Faced Higher Inflation at the Peak



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

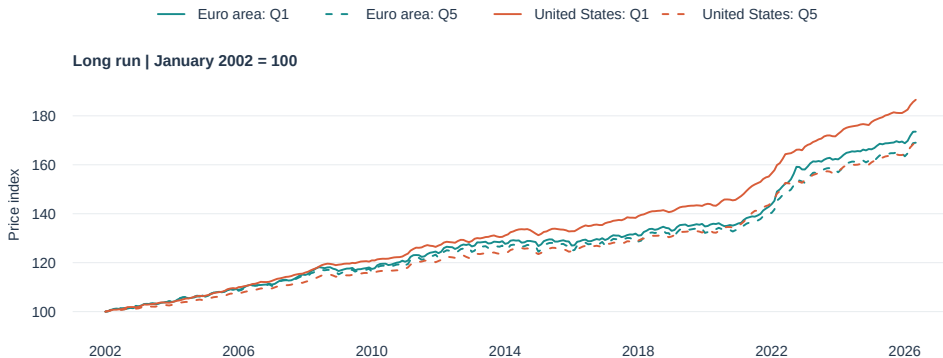
The Post-COVID Shock Widened the Cumulative Price Gap

A return of the annual Q1–Q5 inflation gap to zero does not imply that the cumulative price-level gap has disappeared.



Source: Eurostat HICP and HBS; authors' calculations. Units: price index, 2000=100.

Euro Area and US: Long-Run Inflation by Income

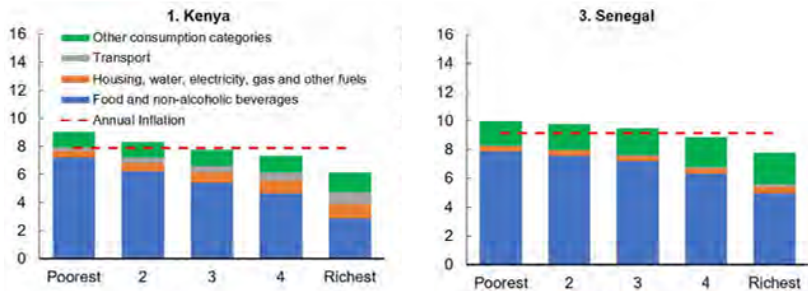


By May 2026, cumulative Q1–Q5 gaps since 2002: **4.5 pp** (EA), **17.3 pp** (US).

Source: Euro area: HICP–HBS excluding rents (041, 042). US: Jaravel (2024), supplied CSV. January 2002 = 100; data through May 2026. Rebasing does not harmonize CPI/HICP coverage or income-group definitions.

Food Prices Raised Inflation More for Poorer Households

Kenya and Senegal, 2021Q2–2022Q2; inflation by quintile (%).

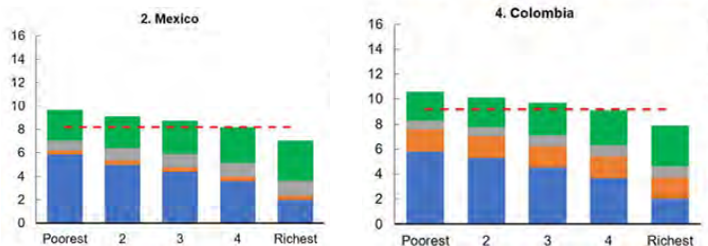


- Faster food-price increases weigh more on poorer households because food absorbs a larger share of their spending.
- In Kenya and Senegal, nominal incomes also failed to keep pace with prices.

Source: IMF, Fiscal Monitor, April 2023, ch. 2, Fig. 2.5 and discussion. Quintiles: per-capita consumption. Dashed lines: national inflation.

Basket Exposure and Income Changes Can Offset Each Other

Mexico and Colombia, 2021Q2–2022Q2; inflation by quintile (%).



- Food also drives the income gradient here, but real incomes rose. Income and wealth effects can outweigh basket exposure.
- Basket gaps can narrow or reverse as relative prices change.

Euro-area benchmark: Q1–Q5 = **1.2 pp in calendar 2022**. Periods and quintiles differ: the IMF uses per-capita income here; our groups use total household income.

Source: IMF, Fiscal Monitor, April 2023, ch. 2, Fig. 2.5 and discussion; euro area: this paper. Colors as on previous slide; dashed lines: national inflation.

Fiscal Measures Directly Reduced Consumer Prices

- **Administered prices:** electricity and gas tariff shields, retail caps and regulated charges.
- **Tax reductions:** temporary VAT and excise cuts on energy and selected food products.
- **Unit support:** fuel rebates and per-kWh subsidies.

Price-based support in the inventory totals **EUR 268.9 billion**: EUR 236.2 billion untar-geted and EUR 32.8 billion targeted.

Reconstructing Prices Without the Measures

For each affected product j and month t , replace its observed price index:

$$\tilde{P}_{j,t} = P_{j,t} \frac{\tilde{p}_{j,t}}{p_{j,t}}.$$

- Use statutory tax rates, unit rebates or documented reference tariffs when available.
- Calibrate an aggregate-equivalent price wedge when monthly tariffs, coverage or unit prices are unavailable.
- Retain observed prices for unaffected products and apply the same household weights and index construction.

France: Defining the Energy-Price Counterfactuals

CF = counterfactual; O = observed; TRV_{gas} : gas tariff (EUR/kWh); TRV_e : electricity tariff (EUR/kWh).

1. Gas Nov. 2021–June 2023

$$\Delta P_{\text{gas},t}^{CF} = \beta_{\text{gas}} \Delta TRV_{\text{gas},t}^{CF}$$

unit-free decimal rates.

$$\beta_{\text{gas}} = 0.82.$$

3. Motor fuels Apr.–Dec. 2022

$$p_{\text{fuel},t}^{CF} = p_{\text{fuel},t}^O + r_t \quad (\text{EUR/litre, incl. taxes})$$

Chained index:

$$P_{\text{fuel},t}^{CF} = P_{\text{fuel},t-1}^{CF} \frac{p_{\text{fuel},t}^O + r_t}{p_{\text{fuel},t-1}^O + r_{t-1}}.$$

Monthly rebates (EUR/litre):

$$\text{Apr.–Aug.} \quad r = 0.18 \quad \text{Sept.–Oct.} \quad r = 0.30$$

$$\text{November} \quad r = 0.22 \quad \text{December} \quad r = 0.10$$

2. Electricity Feb. 2022–Feb. 2024

$$\Delta P_{e,t}^{CF} = \beta_e \Delta TRV_{e,t}^{CF}$$

unit-free decimal rates.

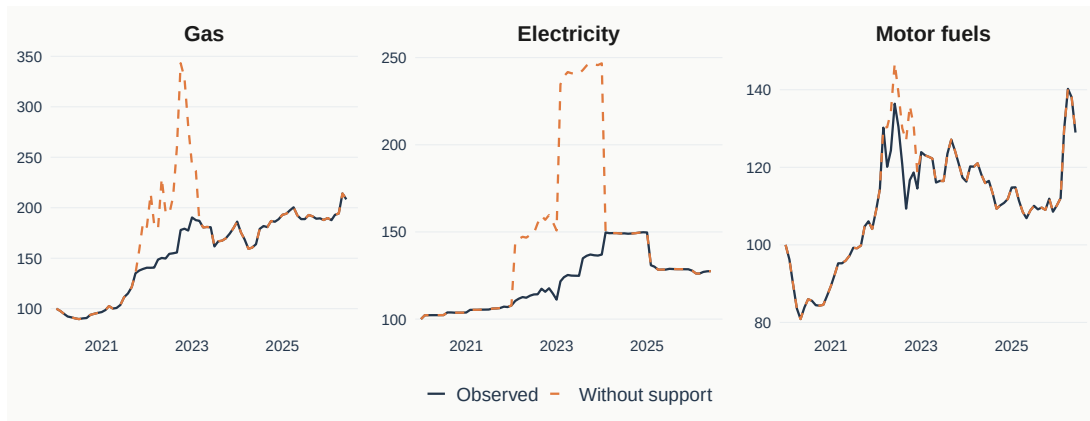
$$\beta_e = 1.02.$$

Date	Observed reference	$\Delta TRV_{e,t}^{CF}$
Feb. 2022	Jan. 2022	33.24%
Feb. 2023	Jan. 2023	108.91%
Feb. 2024	Feb. 2024	0% [†]

[†]Model closure, not a tariff estimate: $P_{e,t}^{CF} = P_{e,t}^O$.

France: Observed and Counterfactual Energy Price Indices

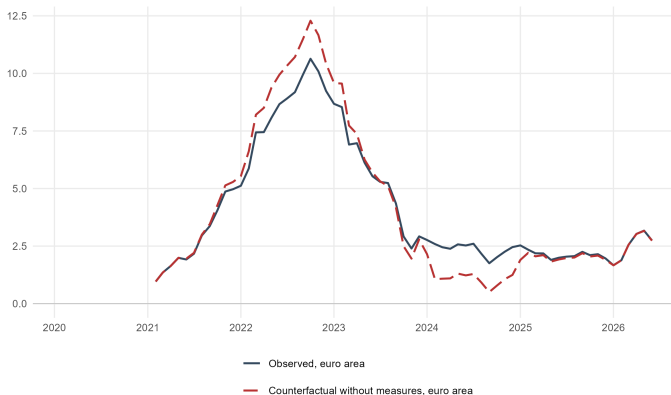
January 2020 = 100; January 2020–June 2026. Vertical scales differ across panels.



Source: French observed and no-policy component price series; authors' calculations.

Price Policies Compressed the Inflation Peak

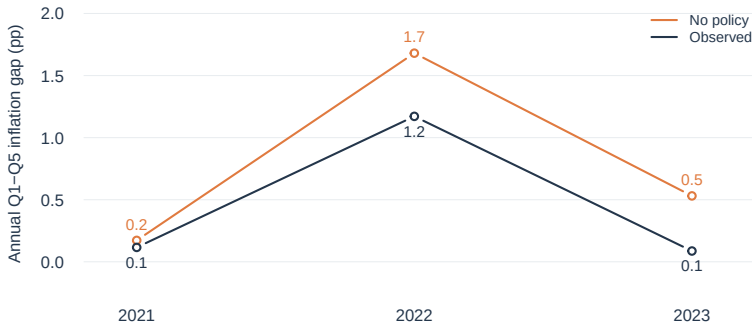
Observed and no-policy year-on-year inflation (%).



Source: Eurostat HICP and HBS; authors' calculations.

Policy Effects on the Q1–Q5 Gap

Price measures can narrow the Q1–Q5 gap without income targeting: electricity and gas support reduced the gap, while fuel support partly offset this effect in 2022.



Source: Eurostat and national statistical institutes; authors' calculations. Annual-average Q1–Q5 inflation-rate gaps, in percentage points.

Household Energy Accounts for Most of the Gap Reduction

Product group	Average-inflation effect			Q1–Q5 inequality effect		
	2021	2022	2023	2021	2022	2023
Food and non-alcoholic beverages	0.0	0.0	-0.1	0.0	0.0	0.0
Housing energy	-0.1	-1.0	-0.6	-0.1	-0.6	-0.4
Vehicle operation	0.0	-0.3	0.3	0.0	0.1	0.0
All other products	0.0	0.0	0.0	0.0	0.0	0.0
Total	-0.1	-1.4	-0.3	-0.1	-0.5	-0.4

- Household energy lowers the Q1–Q5 gap by **0.6 point in 2022** and **0.4 point in 2023**; the 2022 fuel effect offsets 0.1 point.

Source: Eurostat and national statistical institutes; authors' calculations. Product contributions to annual policy effects, in percentage points.

Policy Effects Differed Sharply Across Countries and Years

Country	Average-inflation effect			Q1–Q5 inequality effect		
	2021	2022	2023	2021	2022	2023
Germany	0.0	-0.5	-0.8	0.0	-0.2	-0.8
France	-0.1	-2.4	-0.5	0.0	-1.0	-0.4
Italy	-0.2	-1.4	0.7	-0.1	-0.2	-0.1
Spain	-0.5	-2.1	-0.4	-0.2	-0.8	-0.4
Netherlands	0.0	-1.8	-0.7	0.0	-0.7	-0.7
Portugal	0.0	-0.9	-1.5	0.0	-0.3	-0.9
Slovakia	0.0	0.0	-3.1	0.0	0.0	-2.0
Euro area	-0.1	-1.4	-0.3	-0.1	-0.5	-0.4

- Negative effects indicate that unconventional fiscal policies reduced inflation or the Q1–Q5 gap.

Source: Eurostat and national statistical institutes; authors' calculations. Entries are observed minus no-policy annual-average inflation rates, in percentage points.

Temporary Support Changes the Timing of Inflation

- Lower prices during the policy window compress the inflation peak.
- Withdrawal and comparison with policy-depressed prices a year earlier can subsequently raise measured inflation.
- A positive later inflation effect can therefore coexist with lower prices while support was active.

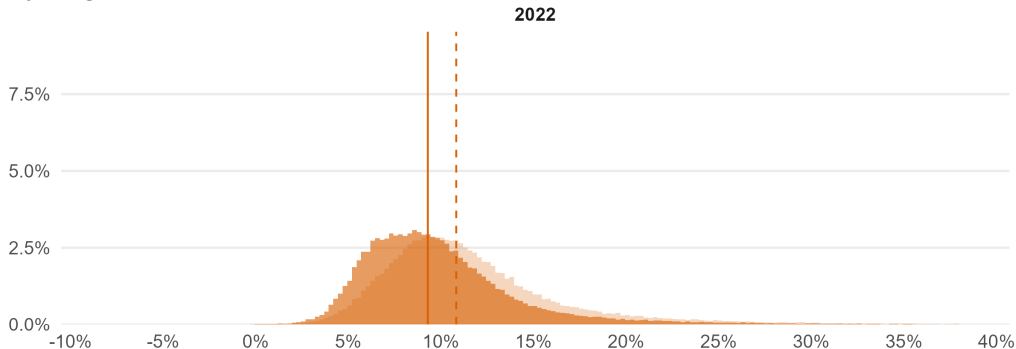
Scope: the counterfactual measures direct consumer-price effects. It does not incorporate fiscal financing, income adjustment, substitution or general-equilibrium responses.

Household Microdata Reveal Individual Exposure

- Combine each household's 2020 expenditure profile with annual HICP weights and common product price indices.
- Household weights use expenditure relative to the national product mean, followed by normalization. These indices do not use RAS.
- Exclude actual and imputed rents. Define national quintiles using total net household income, without equivalisation.
- Construct an index for each household before computing survey-weighted distributions and regressions.

Policies Shifted the Household Inflation Distribution in 2022

Survey-weighted household shares; same baskets in both scenarios.



Observed median: **9.29%**; no-policy median: **10.83%**.

Substantial Dispersion Remains Within Countries

Interquartile range (pp)	2021	2022	2023
Observed household inflation	1.54	4.43	2.62
After country fixed effects	1.29	3.01	1.86
After household controls as well	1.26	2.90	1.83

Household characteristics reduce the remaining within-country IQR by only **3.64% in 2022**.

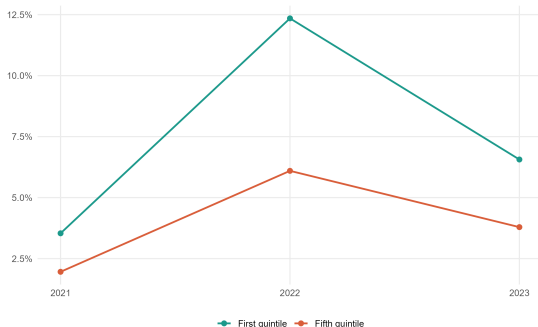
Controls: income quintile, age, household size and residence area. These are changes in interquartile ranges, not additive variance shares.

Income, Age and Residence Gradients Peaked in 2022

Conditional inflation difference (pp)	2021	2022	2023
Q5 relative to Q1	-0.14 (0.21)	-1.50*** (0.40)	-0.11 (0.37)
Under 30 relative to 60+	-0.13 (0.15)	-1.37*** (0.16)	-0.45 (0.34)
Cities relative to rural areas	-0.52*** (0.08)	-1.17*** (0.12)	0.22* (0.12)

Basket-exposure associations conditional on other household characteristics and country effects; 152,440 observations per year.

Inflation Burden Is Larger for Low-Income Households



- The Q1 burden reached **3.5%, 12.3%, and 6.6%** of net income in 2021–2023, versus **2.0%, 6.1%, and 3.8%** for Q5.

Scaled price exposure using 2020 resources, not an observed loss of real income or welfare.

Source: Eurostat HICP and HBS microdata; authors' calculations. 18-country sample; scaled non-rent exposure, in percent of net income. Ratios winsorized at the national 99th percentile.

Appendix

Finer Product Detail Can Widen or Narrow Income Gaps

2022 Q1–Q5 gap (pp)	Macro D3	Micro D2	Micro D3	Micro D4	Micro D5
Germany	1.4	2.0	1.5	–	–
France	0.2	0.3	0.3	–	–
Spain	1.0	0.8	1.0	–	–
Netherlands	2.8	2.0	3.0	2.5	1.9
Belgium	2.3	2.5	2.3	2.4	2.3
Lithuania	3.4	2.9	3.7	4.9	4.9

Finer detail can widen or narrow gaps. Dutch matched coverage: 99.3% for macro D3, 99.9% for micro D3, and 87.8% for micro D5.

Source: Main paper, Aggregation Bias appendix. Selected countries; 18-country table in the paper. Macro: published HBS; micro: reconstructed total-income quintiles. RAS weights, rents excluded. Dashes: unavailable detail.

Prices Paid Within Product Categories Remain Unobserved

Source of heterogeneity	Example	Our indices
Across categories	Different shares of food, energy and services	Captured
Within categories	Different brands, pack sizes or organic vs. regular products	Not observed
Within identical products	Different retailers, promotions or purchase dates	Not observed

Household microdata reveal different baskets; scanner data additionally reveal detailed product choices and transaction prices.

Source: Kaplan and Schulhofer-Wohl (2017), supplied presentation, PDF pp. 2, 9; Jaravel (2022), supplied presentation, PDF pp. 4, 12. Our coverage: main paper.

More Granular Data: Scanner Data and Cheapflation

New datasets are expanding the literature on inflation inequality.

Scanner data: supermarkets, Nielsen, GfK, Kantar.

Advantages	Limitations
Barcode-level prices and quantities	Incomplete coverage: mainly groceries
High-frequency observations	Limited cross-country comparability

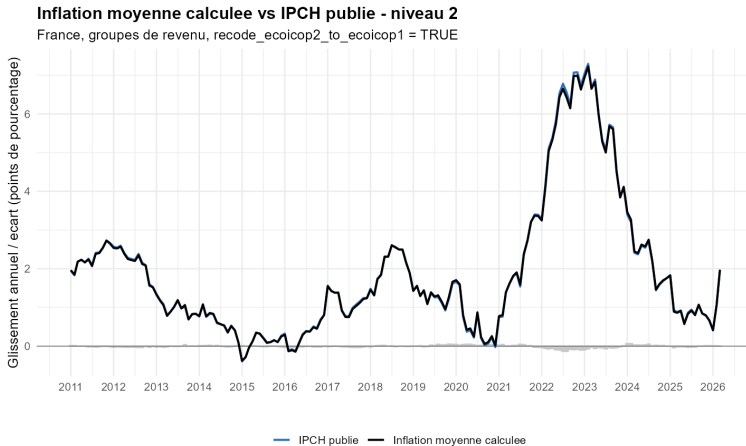
- **More household heterogeneity:** individual indices reveal dispersion within income groups and product categories (e.g., organic vs. regular products).
- **UK evidence, 2021Q3–2023Q3:** cheaper varieties had faster price growth; lower-expenditure households bought more of these varieties.
- **A possible mechanism:** thin margins may limit cost absorption. The same absolute price rise implies higher percentage inflation for cheaper goods.

Source: Chen, Levell and O'Connell (2024), ESCoE webinar, PDF pp. 9–10, 17–19, 26; Sangani (2026), *Cheapflation Cycles*, NBER WP 35235.

Price Measures Cushion Unequal Household Exposure

- Relative-price changes and household baskets jointly determine inflation exposure, with substantial country differences.
- Electricity and gas support account for most of the reduction in the income inflation gap. Fuel support partly offsets it in 2022.
- Across 2021–2023, price policies lower annual inflation by about **0.6 point** and the annual Q1–Q5 gap by **0.3 point** on average.
- Group averages conceal substantial household dispersion and larger exposure relative to income at the bottom of the distribution.

All-Items Reference: RAS Tracks the Published HICP



Source: INSEE, Eurostat; authors' calculations. Original all-items validation; not the excluding-rents baseline.

Year-on-year inflation (%).

Inflation Burden Scales Exposure by Household Resources

For household i in country c , the raw exposure measure is

$$\mathcal{B}_{ic,y}^{\text{raw}} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}.$$

- $\bar{\pi}$: annual household inflation (%); X : total expenditure; Y : net income, both from the 2020 HBS.
- Winsorize expenditure-to-income ratios at the national survey-weighted 99th percentile, then aggregate household burdens.
- Non-rent inflation is scaled by total expenditure: this measures exposure relative to income, rather than an exact welfare loss.

Source: Main paper, Inflation Burden. 18-country sample; Italy and Portugal excluded.

Inflation Inequality Was Highly Uneven Across Countries

Country	2021	2022	2023	Country	2021	2022	2023
Austria	-0.5	0.1	0.5	Ireland	0.5	3.0	1.1
Belgium	1.3	2.3	-2.6	Italy	0.1	1.6	0.5
Croatia	0.1	0.4	-0.3	Latvia	-0.2	4.8	2.5
Cyprus	0.0	0.8	0.2	Lithuania	0.2	3.4	0.4
Estonia	0.5	4.4	1.3	Luxembourg	0.1	0.2	-0.1
Finland	-0.2	-0.9	0.4	Malta	-0.2	0.3	0.5
France	0.3	0.2	0.2	Netherlands	-0.2	2.8	-0.9
Germany	-0.4	1.4	0.6	Portugal	0.8	0.2	-0.9
Greece	0.4	1.4	-0.6	Slovakia	-0.9	1.2	1.4
Spain	0.8	1.0	-0.7	Slovenia	0.1	0.5	1.4
Euro area					0.1	1.2	0.1

Source: Eurostat HICP and HBS; authors' calculations. Units: Q1 minus Q5 annual-average inflation rates, in percentage points.

RAS Calibration: Seed and Margins

HBS expenditure on product j allocated to group g :

$$c_{j,g,y'}^{\text{HBS}} = \frac{E_{j,g,y'}^{\text{HBS}}}{E_{j,\text{all},y'}^{\text{HBS}}}, \quad Z_{j,g,y} = \max\{s_{g,y} \bar{\omega}_{j,y}^{\text{HICP}} c_{j,g,y'}^{\text{HBS}}, 10^{-12}\}.$$

Alternate group and product scaling until both margins match:

$$\widetilde{M}_{j,g,y}^{(t)} = M_{j,g,y}^{(t-1)} \frac{s_{g,y}}{\sum_{\ell} M_{\ell,g,y}^{(t-1)}}, \quad M_{j,g,y}^{(t)} = \widetilde{M}_{j,g,y}^{(t)} \frac{\bar{\omega}_{j,y}^{\text{HICP}}}{\sum_h \widetilde{M}_{j,h,y}^{(t)}}.$$

Return $\omega_{j,g,y} = M_{j,g,y} / s_{g,y}$.

Source: Main paper, RAS Weighting appendix. HBS wave matched to the HICP weight year.

Group and Household Indices Both Exclude Rents

- Exclude actual and imputed rents (HBS 041 and 042) and corresponding HICP rent components before matching and aggregation.
- Reconstruct parent housing categories without rents and normalize weights over matched non-rent products.
- Group indices use RAS; individual indices use relative-expenditure weights. Common coverage does not make the two estimators identical.

Rent sensitivity and official all-items validation retain their explicitly stated reference coverage.

Source: Main paper, Data and Methodology; RAS Weighting appendix.

Household Weights Use Relative Product Expenditure

Let a_{ic} be the survey weight and $e_{ic,j,2020}^{\text{HBS}}$ household expenditure.

$$\bar{e}_{c,j,2020}^{\text{HBS}} = \frac{\sum_i a_{ic} e_{ic,j,2020}^{\text{HBS}}}{\sum_i a_{ic}}, \quad r_{ic,j} = \frac{e_{ic,j,2020}^{\text{HBS}}}{\bar{e}_{c,j,2020}^{\text{HBS}}}.$$

$$\omega_{j,ic,y} = \frac{r_{ic,j} \omega_{j,c,y}^{\text{HICP}}}{\sum_{k \in \mathcal{J}_{c,y}} r_{ic,k} \omega_{k,c,y}^{\text{HICP}}}.$$

$r_{ic,j}$ is expenditure relative to the national product mean, not a household budget share. RAS is not used.

Source: Main paper, household weight equations. Fixed 2020 expenditures, annual HICP weights, matched non-rent products.

Direct and Indirect Exposure to Import-Price Shocks

The observed 2022 extra-EU import-price increases are 83.9% for energy and 21.0% for food.

$$\Delta \mathbf{p}^k = (\mathbf{I} - \mathbf{A}^\top)^{-1} \mathbf{b}^k, \quad \Delta \pi_g^k = \mathbf{w}_g^\top \mathbf{C} \mathbf{d}^k + \mathbf{w}_g^\top \mathbf{C} \Delta \mathbf{p}^k.$$

- $\mathbf{d}^k$: final-import price impulses; $\mathbf{b}^k$: imported-input cost impulses.
- $\mathbf{A}$: FIGARO production network; $\mathbf{C}$: NACE–COICOP mapping; $\mathbf{w}_g$: group expenditure weights.
- Complete pass-through and fixed input coefficients: an accounting exposure simulation, not a causal estimate of observed inflation.

Source: Main paper, Direct and Indirect Exposure to Extra-EU Import-Price Shocks.

Energy Import Prices Widen the Income Inflation Gap

Product group	Average inflation	Q1–Q5 gap	Direct gap	Indirect gap
Food and non-alcoholic beverages	0.6	0.2	0.0	0.2
Alcohol, tobacco and narcotics	0.1	0.1	0.0	0.1
Clothing and footwear	0.1	0.0	0.0	0.0
Housing maintenance and furnishings	0.2	-0.1	0.0	-0.1
Housing energy	1.7	1.2	0.3	0.9
Vehicle operation	1.8	-0.4	0.0	-0.4
Transport (excluding vehicle operation)	1.2	-0.1	0.0	-0.1
Other	0.8	-0.1	0.0	-0.1
Total	6.5	0.7	0.3	0.5

- Housing energy widens the Q1–Q5 gap by **1.2 points**; other products partly offset this effect.

Source: Eurostat trade statistics, HBS and FIGARO; authors' calculations. Observed extra-EU energy import-price shock: 83.9% in 2022. Effects in pp; components sum before rounding.

Pooled Household Inflation Gradients, 2021–2023

Conditional difference (pp)	Coefficient	Standard error
Q5 relative to Q1	-0.582	0.130
Under 30 relative to 60+	-0.651	0.087
Cities relative to rural areas	-0.487	0.036

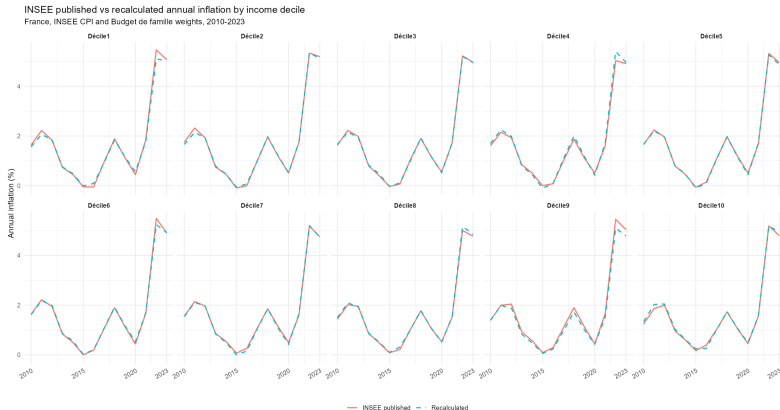
Survey-weighted regression with income, age, residence, household size, and country and year fixed effects. 457,320 household-year observations.

All three coefficients are significant at the 1% level with standard errors clustered by country.

Source: Main paper, pooled household regression, levels specification.

External Validation Against INSEE Income-Decile Inflation

France, annual inflation (%); official and recalculated series.



Source: Eurostat HICP and HBS; authors' calculations. Original reference coverage; not the excluding-rents baseline.

Scanner Data Reveal Dispersion Hidden by Common Prices

United States, Nielsen consumer panel; mean IQR of annual household inflation.

Laspeyres index construction	Mean IQR (pp)
Broad household baskets, CPI prices	1.61
Barcode baskets, national barcode-average prices	3.99
Barcode baskets, household-level prices	7.33

- Detailed product choices and prices paid reveal additional household dispersion.
- These are averages of date-specific IQRs, not additive variance components.

Source: Kaplan and Schulhofer-Wohl, *Inflation at the Household Level*, 31 March 2017 presentation, PDF p. 35.

Dates: 2004Q1–2012Q3. Barcode goods only; not directly comparable with our euro-area 2022 IQR.

A Fixed Basket Measures Exposure, Not Full Welfare Loss

- **Substitution:** households can change products, shops and purchase timing as relative prices change.
- **Living standards:** switching to cheaper varieties when purchasing power falls need not mean that utility is preserved.
- **Product turnover:** entry, exit and quality changes complicate comparisons of prices over time.
- **Our interpretation:** chained indices with fixed HBS profiles measure basket exposure; a welfare calculation requires additional assumptions and data.

Source: Kaplan and Schulhofer-Wohl (2017), supplied presentation, PDF pp. 19–22; Chen, Levell and O’Connell (2024), ESCoE webinar, PDF pp. 20–25; our construction: main paper.

Income and Balance Sheets Can Change Who Loses Most

Study and scope	What basket inflation alone misses
Cardoso et al. (2022) Spain, 2021	Income and nominal-balance-sheet channels exceed relative-consumption effects in absolute size in their exercise.
Pallotti et al. (June 2023) DE, FR, IT, ES; 2021–22	Older households tend to lose more; indebted younger households can benefit from erosion of nominal liabilities. Groups are defined by age and consumption quintile.

Our price indices isolate the consumption-basket channel. A ranking of welfare losses also requires incomes, assets, debt and their adjustment.

Source: Supplied presentations: Cardoso et al., PDF p. 31; Pallotti et al., PDF pp. 4, 12, 34. See also Hobijn (2016), PDF pp. 18–28. Historical study-specific findings, not estimates from this paper.

Granular Data Require Coverage and Matching Checks

- **Matched purchases:** household price changes require observing comparable products at both dates; report matched expenditure and sensitivity to minimum coverage.
- **Panel selection:** continuous recording and attrition can change the households represented; document selection and survey weights.
- **Product definition:** distinguish unit prices, pack sizes and quality. A price-based quality ladder is a proxy, not a direct quality measure.
- **External validity:** scanner goods omit much of services and housing; test aggregate fit before comparing household distributions.

Source: Kaplan and Schulhofer-Wohl (2017), supplied presentation, PDF pp. 5–8, 14; Chen, Levell and O’Connell (2024), ESCoE webinar, PDF pp. 9–10, 17. Suggested diagnostics; not additional tests performed in this paper.

Selected References

Ampudia, M., Ehrmann, M., and Strasser, G. (2024). Shopping behavior and the effect of monetary policy on inflation heterogeneity along the income distribution. *Journal of Monetary Economics*.

Cardoso, M., et al. (2022). The heterogeneous impact of inflation on households' balance sheets. *Banco de España Working Paper*.

Pallotti, F., Paz-Pardo, G., Slacalek, J., Tristani, O., and Violante, G. L. (2024). Who bears the costs of inflation? Euro area households and the 2021–2023 shock. *Journal of Monetary Economics*.

Gautier, E., and Montornès, J. (2024). The Heterogeneous Effects of Inflation in France and in the Eurozone. *Revue d'économie financière*, 153, 193–211.

Kaplan, G., and Schulhofer-Wohl, S. (2017). Inflation at the household level. *Journal of Monetary Economics*.

Jaravel, X. (2024). Distributional Consumer Price Indices. *CEPR Discussion Paper No. 19802*.

Amores, A. F., et al. (2025). Inflation, Fiscal Policy, and Inequality: The Impact of the Post-Pandemic Price Surge and Fiscal Measures on European Households. *Review of Income and Wealth*.

Gautier, E., and Montornès, J. (2026). Inflation Inequality Across the Euro Area. *Forthcoming*.

Question 1: What Generates Inflation Inequality?

Which combination can generate inflation inequality through consumption baskets?

Select one answer.

- A. Different consumption baskets and different price changes across products.
- B. Different consumption baskets when all product prices rise at the same rate.
- C. Differences in wage indexation alone.
- D. Differences in savings and debt alone.

Question 2: What Does the Baseline Method Require?

Which statement does NOT describe the baseline group-level method linking HICP prices to HBS baskets?

Select one answer.

- A. Group averages conceal differences between households within the same income quintile.
- B. Monthly retailer scanner data are required to measure each household's transaction prices.
- C. Household expenditure profiles rely on HBS waves, generally collected every five years.
- D. The baseline abstracts from behavioral responses and substitution between products.

Question 3: Who Faced Higher Inflation in 2022?

During the 2022 energy shock, which groups faced higher inflation in our euro-area results?

Several answers are possible. Select all that apply.

- A. High-income households, compared with low-income households.
- B. Households in cities, compared with rural households.
- C. Rural households, compared with households in cities.
- D. Households with a reference person aged 60+, compared with those under 30.

Question 4: Annual Inflation and Cumulative Price Gaps

Suppose Q1 faced higher inflation than Q5 in 2022, but both groups have the same inflation rate in 2023. What follows?

Select one answer.

- A. The cumulative difference between their price indices disappears in 2023.
- B. Prices must have fallen for Q1 in 2023.
- C. The annual inflation gap is zero, but the relative price-level difference accumulated earlier remains.
- D. Both groups necessarily face the same inflation burden relative to income.